# *Streaming & Video updates*

Advanced Browsing and Messaging Workshop.
Forum Nokia PRO.

Singapore 29th November 2004.



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# Agenda

- Recap of streaming, i.e what it entails.

- 3GPP standards/File Formats in streaming.

- Update of Forum Nokia Tools for Audio/Video conversion.

- Update on devices that support streaming/video.

- Mobile TV and Visual Radio updates.

# What is Streaming?

- Streaming is the transmission of real-time data (commonly audio and video) from a server to a client (terminal), where the client decodes and "plays" the data as it is received.

- Streaming lets the viewer see the content immediately after a short buffering period.

- No data will be stored permanently to the terminating client. (Difference to Downloading & local playback.)

- The end user is not able to forward or send the content to another user. The content can be charged each time it is consumed.

# Business Drivers


- Increased ARPU through traffic and service charges

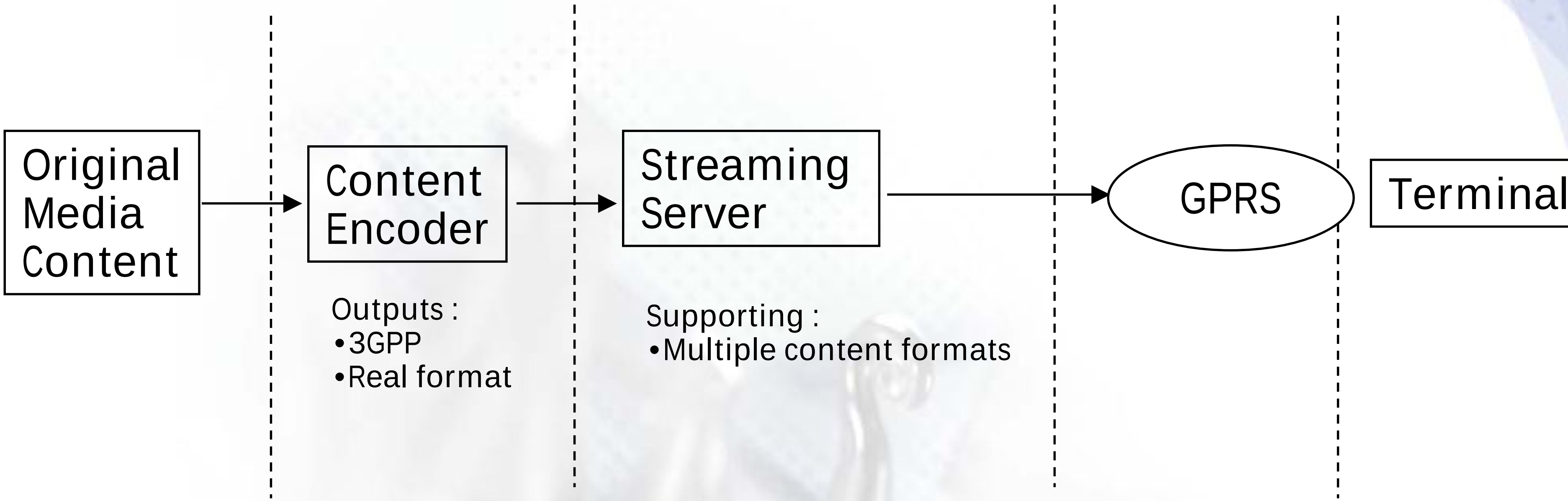
- Video is a high profile and appealing service

- With streaming there is less direct DRM issues involved -> content is never stored on the mobile and cannot be forwarded

- Content is viewed once per streaming session. There is no way to store the content on the terminal -> content can therefore be charged once per session

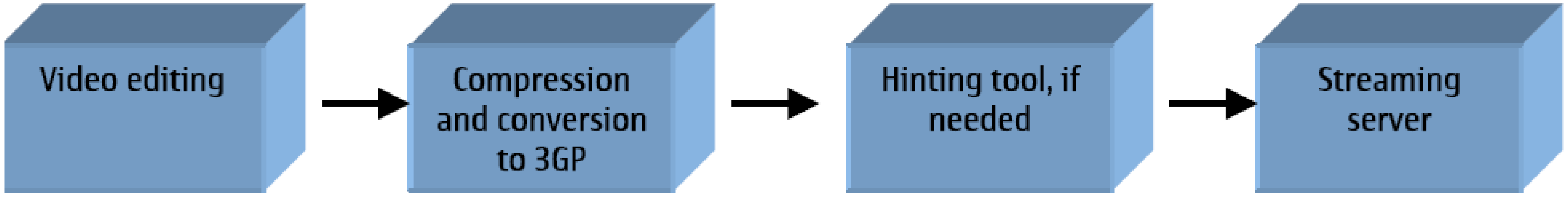
- · The streaming does not consume phone memory -> attractive service for big files and do not compete with downloading

# Streaming Architecture



# Streaming Video creation – Basic Steps

1. Use any of the commonly used video editing tools, e.g., QuickTime, Adobe, etc., to create the video content.
2. Export your content to the compressed format (RealMedia or 3GPP formats) and perform the actual crunching compression.
3. The server extracts the media content from the file and sends it to the network, according to the appropriate payload formats.



# Device Video Features

| | Video recorder | | | | | Video player | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Video codecs | Audio codecs | Image size | Frame rate (fps) | Max. file size, KB | Video codecs | Audio codecs | Image size | Frame rate (fps) | Max. bit rate (Kbit/s) |
| **6170** | H.263 P 0,L 10 | AMR-NB | 128x96 | 15 | 100 or extended | H.263 P 0, L 10, MPEG-4 VSP | AMR-NB | 128x96 | 10-15 | 64 |
| **6230** | H.263 P 0,L 10 | AMR-NB | 128x96 | 15 | 100 or extended | H.263 P 0, L 10, MPEG-4 VSP | AMR-NB | 128x96 | 10-15 | 64 |
| **7200** | H.263 P 0,L 10 | AMR-NB | 128x96 | 7.5 | 100 | H.263 P 0, L 10, MPEG-4 VSP | AMR-NB | 128x96 | 10-15 | 64 |
| **6630** | H.263 P 0,L 10 | AMR-NB | 176 x 144 | 15 | 100 or extended | H.263 P 0, L 10, MPEG-4 VSP Real 7/8 | AMR-NB RA | 176 x 144 | 10-15 | 128 |
| **7710** | H.263 P 0,L 10 | AMR-NB | 176 x 144 | 15 | 100 or extended | H.263 P 0, L 10, MPEG-4 VSP Real 8 | AMR-NB RA | 176 x 144 | 10-15 | 128 |



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# Streaming

3GPP Streaming over RTP

Real Networks proprietary

Real Networks
Helix Mobile Server

Apple: Darwin
Philips: Platform 4

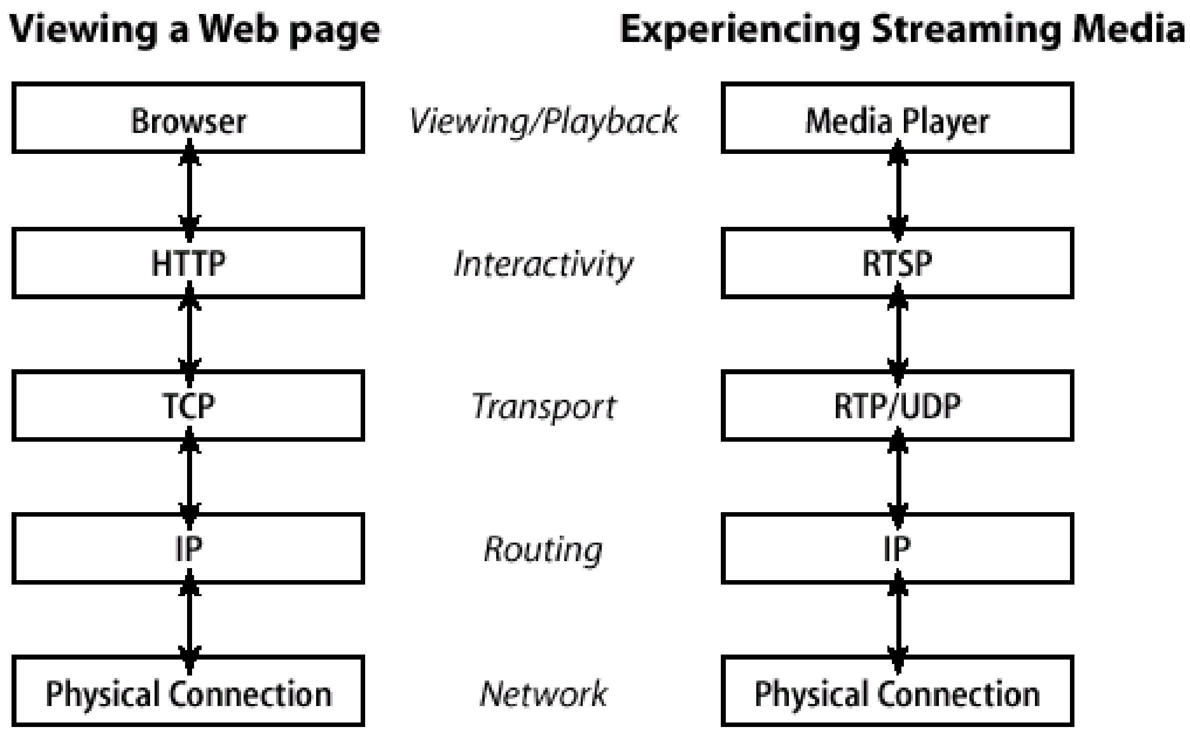
Real Networks
Helix Universal Server

Clients:
Series 40, Series 60, Series 80 and Nokia 7710

Clients:
Series 60, Series 80 and Nokia 7710

# Streaming Network Model

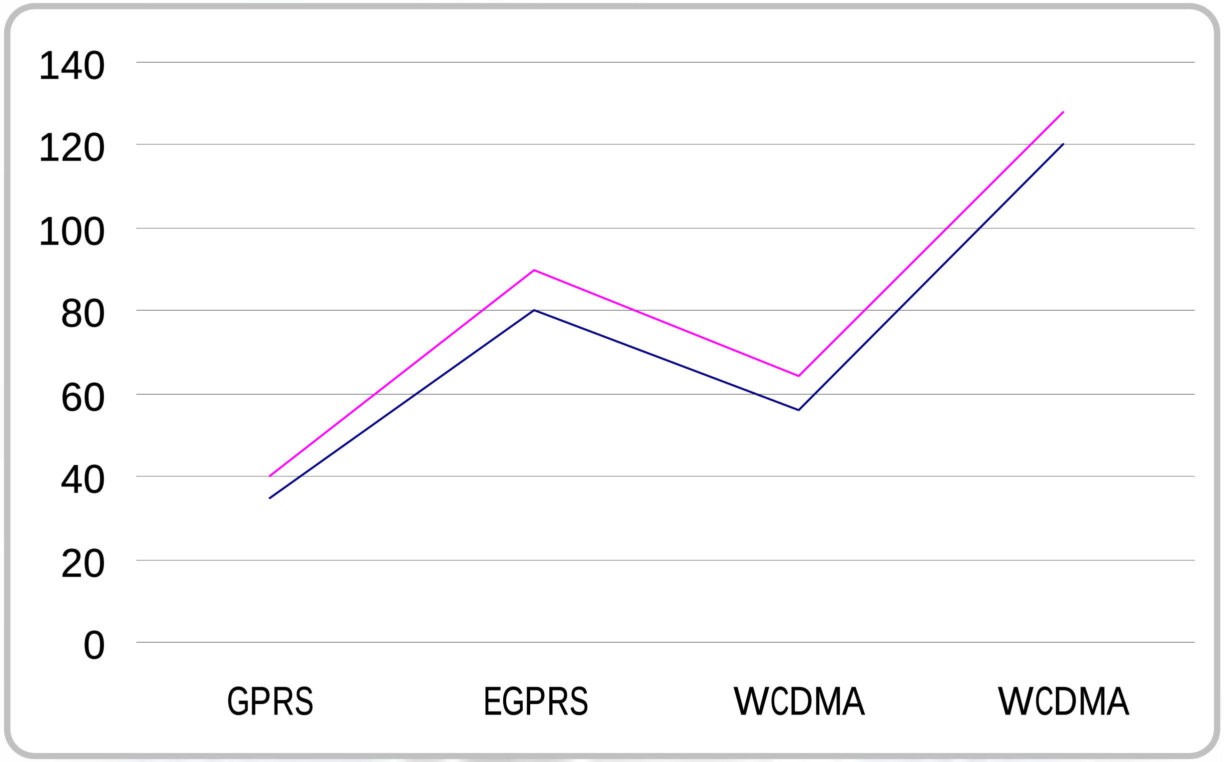
*This very simplified chart shows how protocols are layered one on top of another and, on the protocol level, how the system for viewing a Web page relates to the system for experiencing streaming media.*



# Typical video services

| | Short P2P MMS | MMS video clips | Video (XHTML Downloads) | Video Streaming |
|---|---|---|---|---|
| Typical Initiation | P2P (Person to Person) | A2P (content push/pull to person) | Pull, A2P | Pull, A2P |
| Payer / Price | Calling Party Pays (P2P) | Requestor pays | Requestor (Receiver) Pays (Content) | Receiver Pays (Content) |
| Characteristics | User-generated content | Promo-clips, Small files only (<15 sec), | Larger files | Complement MMS & DL<br>One time view,<br>On Demand & Live stream |
| Typical service | Self-recorded short P2P video | Short A2P video/audio clips, Promotions, messaging etc. | Music Video, Movie & Game preview, branded video clips for msg etc. | News on Demand<br>Full music videos, dedicate streams, live events (horse-racing) |
| Typical File Size | < 100 KB (< 15 sec) | < 100 KB (< 15 sec) | > 100 KB (~300KB) | Large files<br>LIVE broadcast |

# Bit rates for streaming (kbit/s)



# 3GPP

# 3GPP Specification description

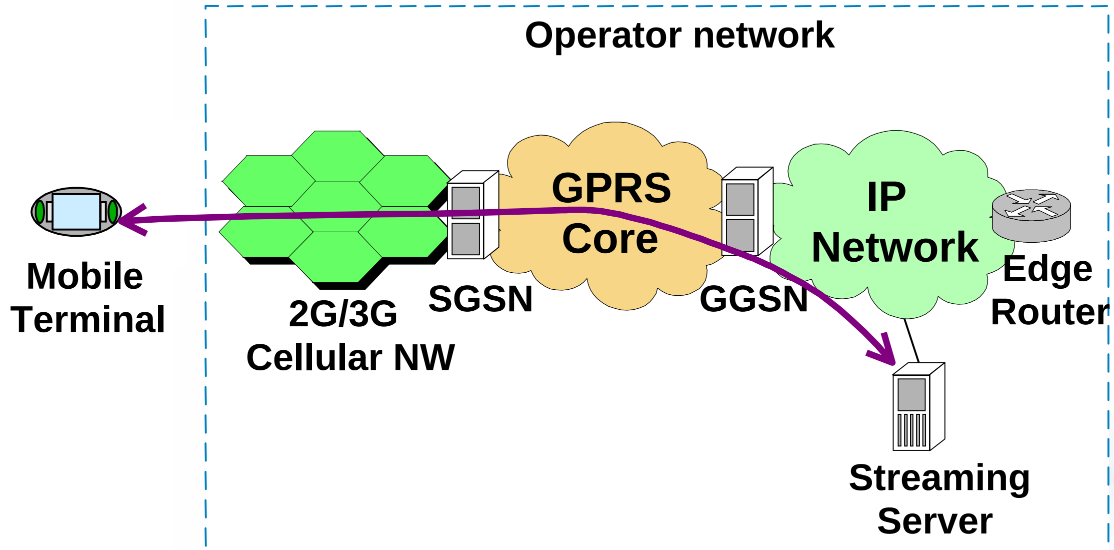
| Specification | Description |
|---|---|
| TS 26.233: Transparent End-to-End Packet-switched streaming Services ( PSS ); General Description | A General description of a transparent packet switched streaming service in 3G networks |
| TS 26.234: Transparent End-To-End Packet-Switched Streaming services ( PSS ); Protocols and Codecs | Specifies the protocols and codes for the streaming service, standardized and deployed within 3GPP compliant systems. |

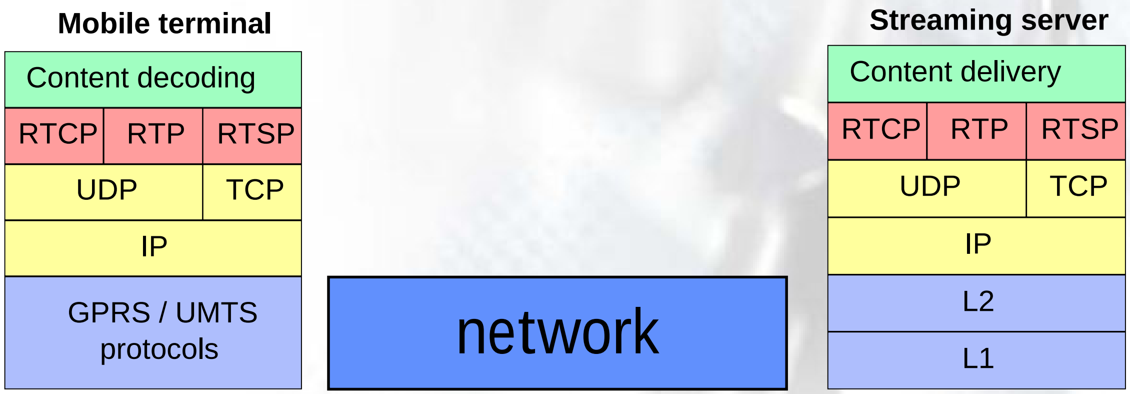
# Evolution to 3GPP PSS Standards

| Today | 3GPP |
|---|---|
| .ram-file format | .sdp-file format |
| Real Networks media formats | 3GPP media and file formats |
| RDT as transport protocol | RTP as transport protocol |
| Real Network Media Producer | 3GPP media producer |

Server and Gateway Infra of today is future proof, so can be re-used in 3GPP

# 3GPP standardised streaming protocols




**3GPP streaming standards**

- 3GPP adopted IETF protocols
  - RTSP+SDP / TCP / IP == session control
  - RTP+RTCP / UDP / IP == data transfer
- 3GPP defined video codecs :
  - H263 Baseline
  - H263 Profile 3 Level 10 (Optional)
  - MPEG-4 (Optional)
- 3GPP defined audio codecs :
  - MPEG-4 AAC (optional)
- 3GPP defined speech codecs :
  - AMR narrow-band (4.75 - 12.2 kbps)
  - AMR wideband (6.60 - 23.85 kbps)

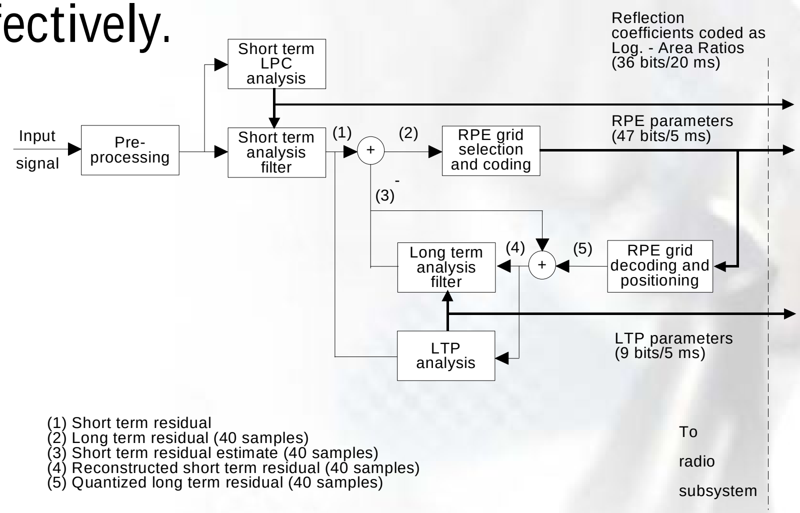
# Why 3GPP PS Streaming

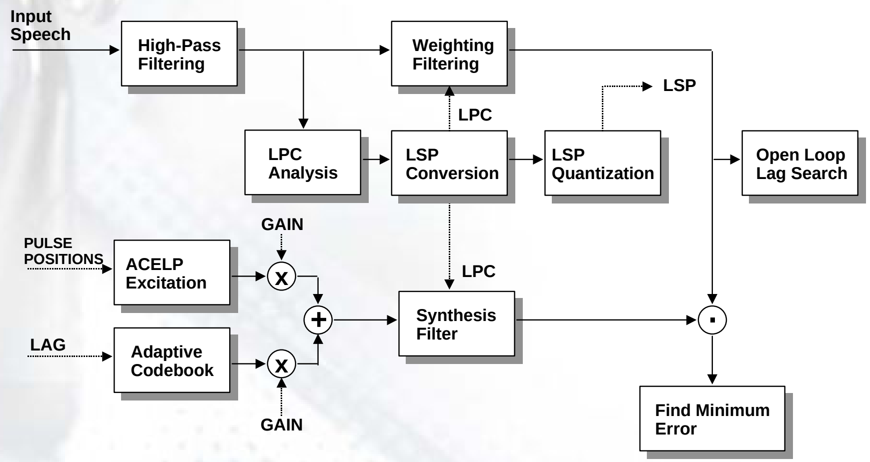
- Avoids fragmentation of services and platform
  – mutually beneficial
- Wider choice of mobile optimised content
  – easier integration with 3GPP streaming services
  -eg. streaming with MMS only interwork with specified 3GPP codecs
- Performance optimizations
  - ensures compliance with network development
- Best consumer experience
- IOT cooperation between Nokia, Ericsson, Real, PV, Emblaze on 3GPP PSS in IMTC (Int'l Multimedia Telcom Consortium) between diff infra vendors
- Support from major professional content tool creation/playback of 3GPP content – Philips, Real, Quicktime, Adobe, Avid, Discreet, Sonic Foundry, Voice Age, Canopis

# Audio Codecs

# Speech Coding

- Speech coding is a means of compressing voice in the mobile, before transmitting it to the receiving end. Compression is achieved by modeling the voice characteristics mathematically.
- The idea is to transmit only the essential information i.e. information needed to be able to reproduce (decode, decompress) the coded voice at the receiving end.
- The benefit is that the mobile network capacity is used more effectively.



# Speech Coding

The general rule is, that the lower the coding bit rate is, the worse the voice quality is. However, using advanced and complex speech codecs it is possible to improve the voice quality. E.g. enhanced full rate codec and full rate codec has the same bit rate but because enhanced full rate uses a more advanced coding technology, it clearly gives improved speech quality than full rate does.

The coded speech needs to be protected against errors in the air interface. How much the error correction is used also effects the voice quality i.e. the more added error correction (channel coding) the better the codec performs in bad channel conditions.

The total bit rate used for voice connections in the GSM system is divided by speech codec and error correction/control. Therefore, if you want to increase the speech coding bit rate, you must use less error correction, and the performance in bad channel conditions will get worse.



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# AMR Audio Codec

- Adaptive Multi Rate (AMR) is the fourth speech codec defined for the GSM system and represents the future of speech coding.
- When the AMR codec was specified, the goal was to combine the benefits of EFR and HR codecs: improved voice quality and improved capacity.

Source: Rysavy Research “Voice Capacity Enhancements for GSM Evolution to UTMS”, p. 5-6

# Introducing AMR

- Currently Nokia GSM/GPRS/EDGE handsets support one of the below combinations:
  - EFR (Enhanced full rate) and HR (Half rate) codecs
  - EFR (Enhanced full rate), HR (Half rate) and AMR (Adaptive Multirate) codecs
  - HR will be discontinued gradually during the course of 2005-2006, i.e
    - only EFR and AMR will be supported in those handsets
- In WCDMA , AMR is the only voice codec supported by the standard, so a WCDMA handsets support AMR in WCDMA.

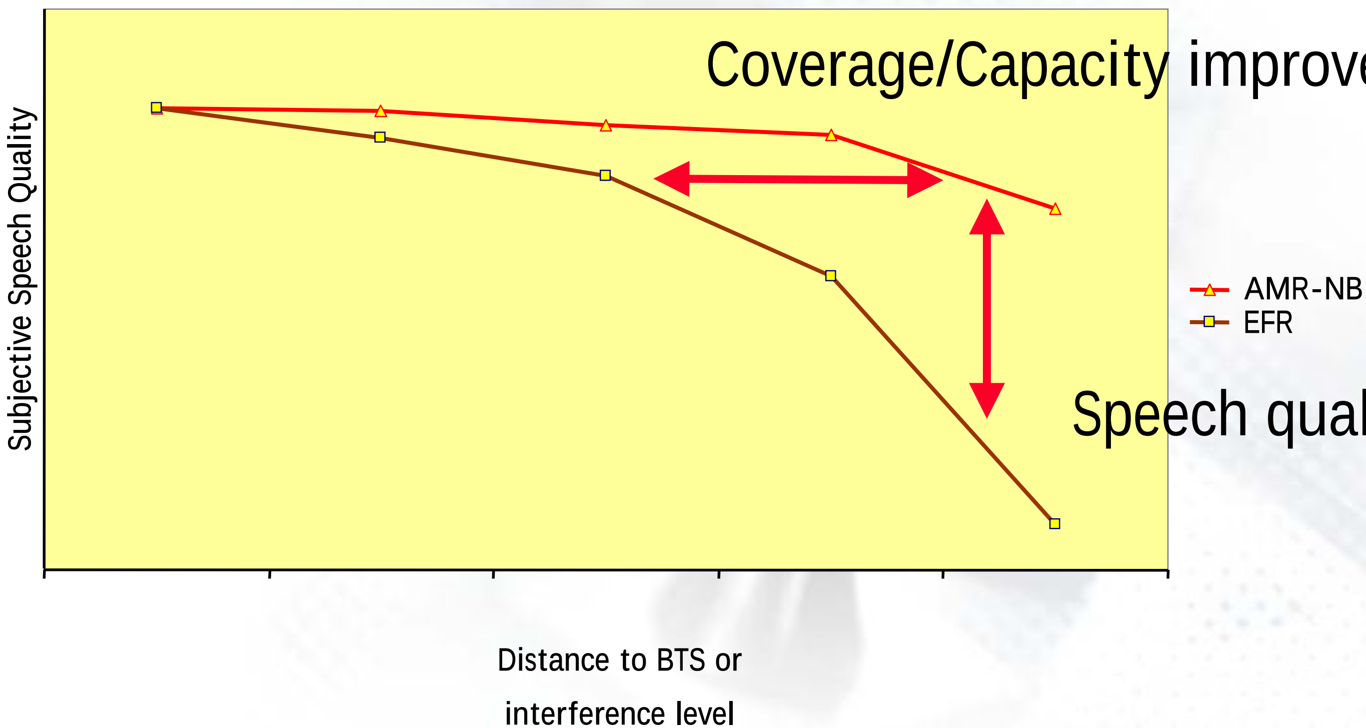
# AMR Basics

AMR (maximum code rate FR 1/5) is more robust than

TCH/FR, TCH/HR and TCH/EFR (maximum code rate 1/2)

⇒ improved speech quality in coverage or interference limited areas

Subjective Speech Quality Degradation in Function of Channel Quality
(AMR-NB vs. EFR)

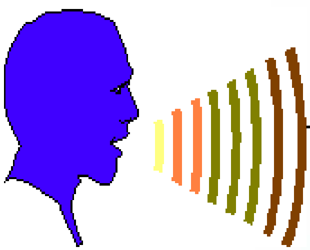



The use of AMR increases substantially the GSM network capacity for the same quality criteria

AMR HR enables higher use of HR speech thereby improving the capacity

# Four different Speech Codecs defined for the GSM system

**FR = Full Rate**
**HR = Half Rate**
**EFR = Enhanced Full Rate**
**AMR = Adaptive Multi Rate**



64 kbps

**13 kbps: FR Speech Codec** → **22.8 kbps**

**5.6 kbps: HR Speech Codec** → **11.4 kbps**

**13 kbps: EFR Speech Codec** → **22.8 kbps**

**4.75 - 12.2 kbps: AMR Speech Codec** → **11.4 or 22.8 kb**

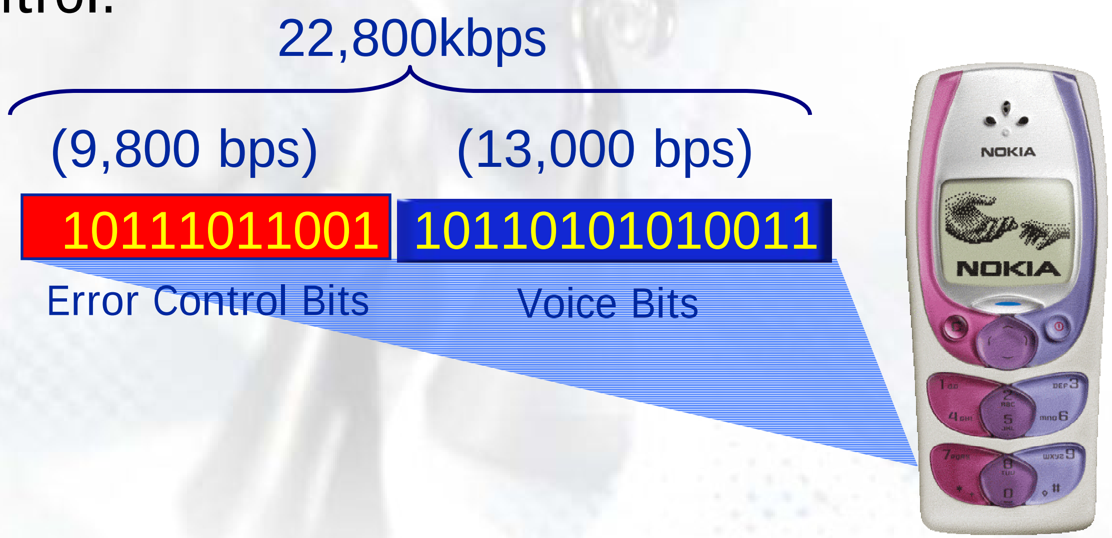
Bit rate before GSM speech coding

Bit rates after speech coding

Bit rates after Channel codin

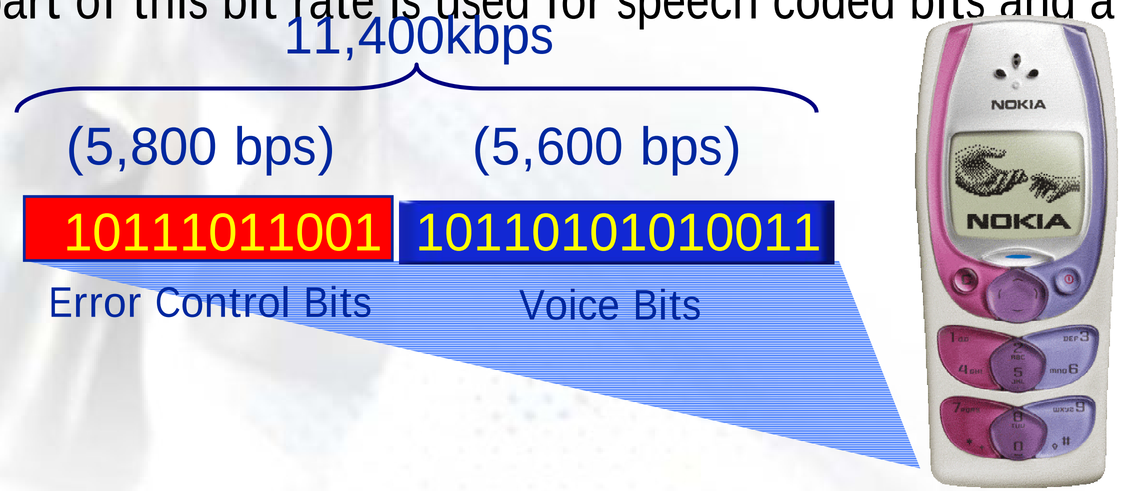
# Full Rate (FR) Speech Connection

- Full Rate Codec is the first speech codec defined for the GSM system.
- FR is mandatory for all GSM mobiles, and has been supported in all GSM mobiles from the start of the GSM system in 1992.
- FR uses one GSM full rate traffic channel with a gross bit rate of 22.8 kbps. A part of this bit rate is used for speech coded bits and a part for error control:



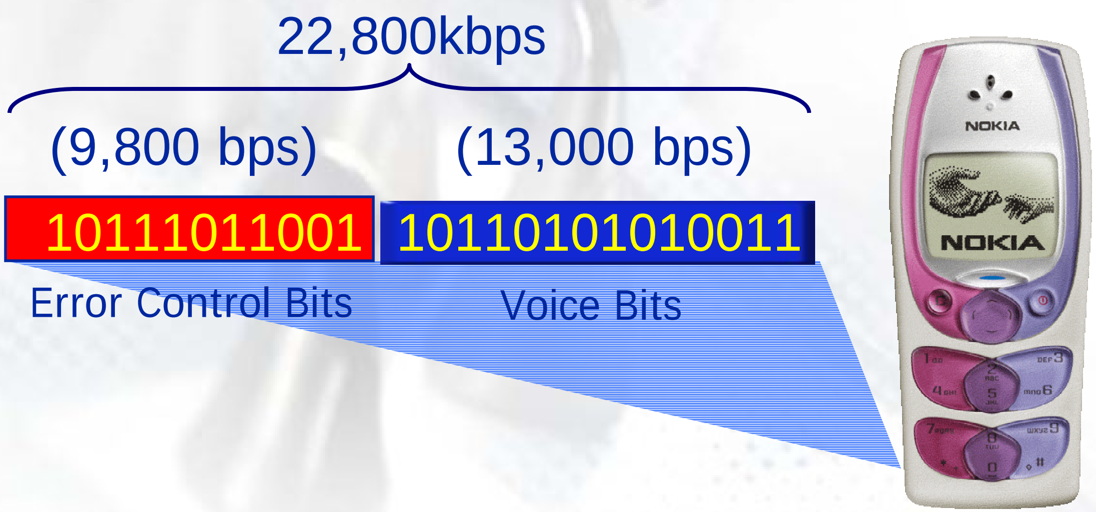
# Half Rate (HR) Speech Connection

- Half Rate Codec is the second speech codec defined for the GSM system.
- HR codec was specified to add more capacity to the GSM system. This was achieved by using half rate traffic channels and lower bit rate speech codec. Through this, the voice capacity of GSM system could be doubled in best case scenario. The first mobile supporting HR was introduced in 1997.
- HR codec uses a more advanced coding method then the one used with FR. Therefore it gives nearly the same speech quality as with FR.
- It uses one GSM half rate traffic channel with a gross bit rate of 11.4 kbps (half of the full rate traffic channel). A part of this bit rate is used for speech coded bits and a part for error control:



# Enhanced Full Rate (EFR) Speech Connectio

- Full Rate is the third speech codec defined by the GSM system.
- The EFR codec was specified to improve the voice quality of the GSM system. EFR gives a clearly better voice quality, compared to FR or HR codecs. This was achieved by using a more complex and advanced speech coding algorithm, than the one used with FR. The first mobile supporting EFR was introduced in 1997.
- The same as with the full rate codec, it uses one GSM full rate traffic channel with a gross bit rate of 22.8 kbps. A part of this bit rate is used for speech coded bits and a part for error control:



# Adaptive Multi Rate (AMR) Speech Connecti

Adaptive Multi Rate (AMR) is the fourth speech codec defined for the GSM system, and it represents the future of speech coding.

The goal, when specifying the AMR codec was to combine the benefits of EFR and HR codecs: an improved voice quality and an improved capacity.

AMR achieves this goal by dynamically adapting its bit-rate allocation, between speech and channel coding, thereby optimizing the speech quality in various radio channel conditions: when the channel conditions are good, less bits are used for error correction and more for voice coding, and when the channel conditions are bad more bits are used for error correction and less for voice coding.

Depending on the radio channel conditions, AMR dynamically uses either the GSM full rate traffic channel with a gross bit rate of 22.8 kbps or the GSM half rate traffic channel with a gross bit rate of 11.4 kbps. A part of this bit rate is used for speech coded bits and a part for error control:

11,400 or 22,800kbps

| (3,610 - 10,600 bps) | (4,750 - 12,2 bps) |
|---|---|
| 10111011001 | 10110101010011 |
| Error Control Bits | Voice Bits |

# What is AMR codec

- AMR (Adaptive Multi-Rate) is a digital voice codec containing 8 possible data rate steps: 4.75, 5.15, 5.9, 6.7, 7.4, 7.95, 10.2 and 12.2 kbit/s.
  - AMR 4.75 kbit/s gives the same capacity as GSM half rate codec, so that is sometimes called as “AMR half rate”
  - AMR 12.2 kbit/s gives the same capacity as GSM EFR codec, so that is sometimes called as “AMR full rate”.

# Benefits of AMR

The AMR benefits :

- Clearly better speech quality than EFR or half rate
- Improves coverage
- Enables consistent voice quality between GSM and WCDMA networks
- Maintains the possibility of HW savings in the similar way as Half rate
- Maintains the possibility of capacity improvement in the similar way as Half rate

# Benefits of AMR : speech quality

There is clear difference between speech quality between AMR and half rate. MOS (Mean opinion score, with scale 1 to 5) is typically used for evaluating codecs

- AMR has 4.0 MOS (=good) at its best
- Half rate has 3.2 MOS (=satisfactory) at its best

- Since MOS < 3.2 is already considered relatively bad speech quality, there is more margin against high load when utilising AMR.

Amount of traffic is increasing in networks

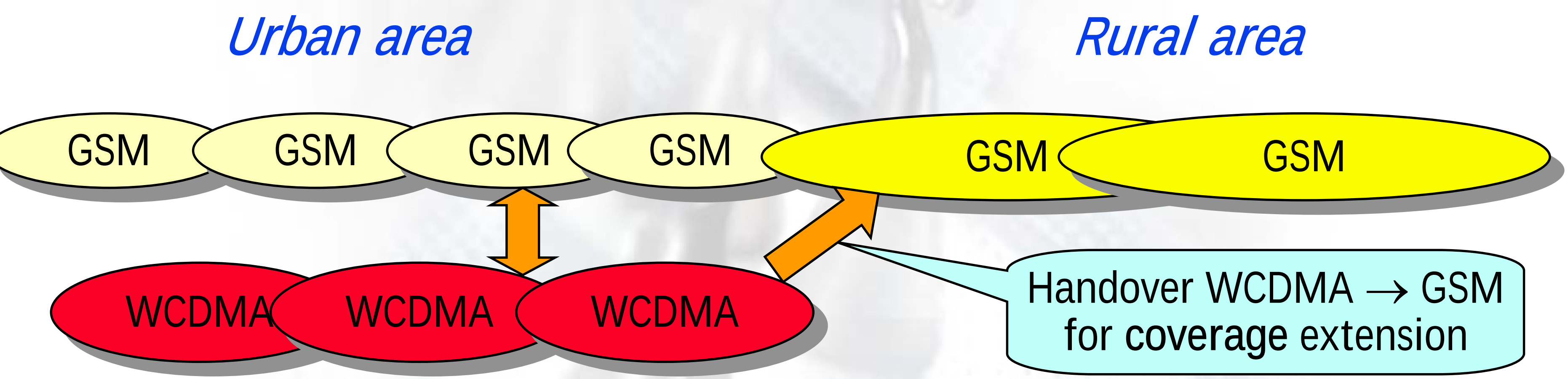
- Amount of data services is increasing in the networks. More traffic will result in more interference, which will result in further decrease in CS voice quality, but end users might not tolerate deterioration of voice quality compared to what they are used to.

# Reason of better voice quality due to AMR

- AMR adapts to different load situations very fast ,
  - uses the smaller codec rate *only when needed* (high load situation).
  - Thus lower (lower than MOS 4.0) speech quality is typically used only for relatively small duration of the phone call and will not be noticed.
- With half rate codec,
  - the codec rate is static *during the whole speech call*.
  - then sub-optimum (MOS 3.2) speech quality is used during the *whole* speech call even without any sudden interference situations.
  - Also, if high load occurs, MOS is further degraded below MOS <3.2 , non-satisfactory level.

# AMR enables consistent voice quality between GSM and WCDMA networks

- At least in initial phase, WCDMA is typically not available everywhere, so handovers between GSM and WCDMA are needed , when users move around .
- It is only AMR codecs which are used in WCDMA network,
  - Consistent voice quality is desired from dual mode users
  - If AMR is used also in GSM network, there is seamless voice quality also after handovers.



# Adaptive Multi Rate (AMR)

- Adaptive Multi-Rate Narrowband Speech codec ( AMR-NB )

- Mandatory speech codec in 3G PSS
- Mainly used for speech encoding at very low bit rate
- Sampling frequency : 8000 HZ mono
- One AMR frame = 20 ms = 160 samples
- Can be use for both constant bit rate ( CBR ) and Variable bit rate ( VBR ) encoding/decoding
- VBR can be achieved by dynamically changing the bit-rate switch, switching can be done on a frame by frame basis.

# Adaptive Multi Rate (AMR)

- **Adaptive Multi-Rate Wideband speech codec ( AMR-WB )**

- Mainly used for high-quality speech/low quality audio encoding at low bit rate
- Sampling frequency : 16000 hz, mono
- One AMR-WB frame = 20 ms = 320 samples
- Switching can be done on a frame by frame basis
- Can be used for both Constant Bit Rate ( CBR ) and Variable Bit Rate ( VBR ) encoding/decoding

# MPEG-4 Advanced Audio Codec ( AAC )

- Optional audio codec in 3G PSS
- Mainly used for high-quality audio encoding at reasonable bit rates
- Sampling frequency: Up to  48000 hz, mono/stereo
- Support for AAC-LC ( low complexity mode )
- One AAC frame = 1024 samples
- Can be used for Constant Bit Rate ( CBR ) and Variable Bit Rate ( VBR ) encoding/decoding

# Video Codecs

# Video – H.263

H.263

- Mandatory Video codec in 3G PSS
- Supports H.263 Profile 0 Level 10; may also support Profile 3 level 10
- Frame size : QCIF ( 176 * 144 ) or Sub-QCIF ( 128  * 96 )
- Frame rate: Max 15 FPS
- Bit Rate: Max 64 kbits /s*
- Content can be encoded using constant frame rate or variable frame rate, not exceeded the selected frame rate

# Video – MPEG4 Visual

- Optional video codec in 3G PSS
- Support Visual Simple Profile Level 0
- Frame Size:  Max QCIF (176 * 144 ) Suggested QCIF or Sub-Qcif
- Frame rate : Max 15 FPS
- Bit rate Max 64 kbits
- Content can be encoded using constant frame rate or variable frame rate, not exceeding the selected frame rate.

# Video – Suggested Video Encoder Configurations

| Video Bit Rate ( Only Video, Kbits ) | Frame size ( Pixels ) | Suggested Intra Refresh Period Range ( Seconds ) | Suggested Frame rates ( Constant FPS ) |
|---|---|---|---|
| 15 | Sub-Qcif ( 128 * 96 ) | 5 – 10 | 5 to 7.5 |
| | QCIF ( 176 * 144 ) | 5 -10 | 2.5 to 3 |
| 20 | Sub-Qcif ( 128 * 96 ) | 5 -10 | 7.5 to 10 |
| | QCIF ( 176 * 144 ) | 5- 10 | 3 to 5 |
| 25 | Sub-QCIF ( 128 * 96 ) | 5 -10 | 10 to 15 |
| | QCIF ( 176 * 144 ) | 5 – 10 | 5 to 7.5 |
| 30 | Sub-QCIF ( 128 * 96 ) | 5 -10 | 10 to 15 |
| | QCIF ( 176 * 144 ) | 5 -10 | 7.5 to 10 |
| 35 | Sub-QCIF | 5 – 10 | 10 to 15 |

# 3GPP release 6 protocols and codecs

- Session establishment

  - Client shall support initial session descriptions specified in one of the following formats: SMIL, SDP, or plain RTSP URL.

  - In addition the following are also valid file:// (for locally stored files) and http:// (for presentation descriptions or scene descriptions delivered via HTTP).

# 3GPP compatible authoring tools

| | Video Conversion | Input | Output | Basic Editing ( Cut and Paste ) | Hint Track generation | Server |
|---|---|---|---|---|---|---|
| Real Networks | Helix Producer | *.AVI, *.MPEG, *.etc | *.rm | No | Not needed for Real Streaming | Yes for real media |
| Real Networks and Envivio | Helix mobile producer powered by envivio | *.AVI, *.MOV, *.MPEG, *.etc | *.3GP | No | Yes for 3gpp serving real servers | Yes, Helix universal serer |
| Nokia | Nokia Multimedia converter 2.0 | *.AVI, *.MOV, *.MPEG | *.3GP | No | No | Non-Commercial Test server |
| Philips | Platform 4 Encoder | *.AVI, *.MPEG | *.3GP | Limited | Yes, for philips server | Yes |
| Apple | Quicktime 6.3 Pro | .AVI, .MOV, .MPEG, etc | *.3GP | Yes | Yes, for apple darwin server | Yes, darwin "open source server" |
| PacketVideo | PV Author | .AVI, .MOV, .MPEG, . | *.3gP | No | | PV Server v4.x |

# Example : Real Networks Streaming solutions

# Helix Mobile Producer

- Using a low quality compressed source will yield a lower quality encoded result
  - Artifacts that exist in the source file will also be carried over to the encoded output.
  - Encoder forced to waste bits by duplicating undesirable artifacts

- Best possible results :
  - Encoding from one of the following formats
    - Uncompressed Quicktime ( .mov )
    - Uncompressed AVI
    -
  - Audio, use uncompressed 44.1 kHz 16 bit sources

- Capturing from devices
  - Good DV camera connected to an Osprey or other mid-range to high-end capture card
  - E.g Osprey 100 ( mid range ), Osprey 500 DV ( High-End )

# Transport Protocols - TCP

## Transmission Control Protocol (TCP)

- Helix Universal Server can use TCP in a number of ways. Because TCP offers a single channel for bi-directional communication, Helix Universal Server uses it as a control channel to communicate with clients about passwords and user commands such as pause and fast-forward. The TCP protocol also guarantees delivery of packets, and has built-in congestion control that helps to provide reliable communication.
- On the down side, TCP responds slowly to changing network conditions, and creates network overhead through its error checking facility. For this reason, TCP is best suited for delivering low-bandwidth material like passwords or user commands. In some cases, TCP can facilitate communication through a firewall. For example, firewalls that block UDP traffic between Helix Universal Server and its clients may permit TCP connections.

# Transport Protocol - UDP

## User Datagram Protocol (UDP)

- Helix Universal Server uses UDP packets to deliver data to client software on its data channel. Client software sends UDP-based requests to Helix Universal Server when packets on the data channel have not arrived. Because the transport does not consume as much network overhead, it can deliver packets faster than TCP.
- Because video and audio data typically consume large amounts of bandwidth, it makes the most sense to use UDP to deliver streaming media. For this reason, Helix Universal Server uses UDP as the default for transmitter-to- receiver communication. With Helix Producer, you can set up the encoder to act as a transmitter. You therefore can, and should, use a UDP connection to deliver live feeds from the encoder.

# Real-Time Streaming Protocol (RTSP)

- standards-based protocol for multimedia presentations,
- very useful for large-scale broadcasting.
- Only RTSP can deliver SureStream files with multiple bit-rate encoding. SMIL, RealText, and RealPix also require RTSP.
- RTSP uses TCP for player control messages, and UDP for video and audio data.
- Use RTSP with RTSP-compatible players such as RealOne Player, RealPlayer, and QuickTime Player.

# Packet Formats

- Helix Universal Server can use both of the following packet formats.

## RealNetworks Data Transport (RDT)

- When Helix Universal Server communicates to a RealNetworks client such as RealOne Player over RTSP, it uses RDT as the packet format.
- proprietary format

## Real-Time Transport Protocol (RTP)

- standards-based packet format designed as the companion to the RTSP protocol.
- QuickTime Player, for example, uses RTP as its packet format. Helix Universal Server fully supports RTP, and shifts to RTP automatically when streaming to an RTP-based client such as QuickTime Player. RealNetworks clients such as RealOne Player also support RTP, using this format when receiving data from RTSP/RTP servers.

# Recommended 3GPP settings for Wireless networks

| Network | Total Bit Rate | Voice Audio | Music Audio | Video | FPS |
|---|---|---|---|---|---|
| GPRS | 15-25 kbps | AMR-NB<br>4750 – 7950 bps | AAC-LC<br>8 kbps | H.263 or MPEG4<br>10-20kbps<br>SQCIF or QCIF | 5-6 |
| UMTS | 50 kbps | AMR-NB | AAC-LC<br>12-16 kbps | H.263 or MPEG4<br>39.8kbps *voice | 6015 |

# Recommended 3GPP settings for Local Playback

| Device | Total Bit Rate | Voice Audio | Music Audio | Video | FPS |
|---|---|---|---|---|---|
| Nokia series 60 | 34-80kbps | AMR-NB<br>12200bps | AMR-NB<br>12200bps | H.263, 22-68kbps<br>QCIF | 5-8 |
| PocketPC | 34-200kbps | AMR-NB 12200bps | AAC-LC 16-32kbps | H.263 22-68kbps | 5-8 |
| Motorola,<br>Sony Ericsson | 34-80 kbps | AMR-NB 12200bps | AAC-LC 16-32 kbps | H.263, 22 – 68 kbps QCIF | 5-8 |

# Mobile Video study

# Mobile Video Services – Use Case Segmentation

| | Non-Real Time | Real Time |
|---|---|---|
| rson-to-Person 2P) | • Video MMS | • 1 or 2-Way Video Call |
| ntent-to-Person 2P) | • MMS push<br>• Browse & download | • Streaming |

# Qualitative Research Conclusions

- 2-way video communication lacks motivation for the majority of respondents.
- Little interest in being able to ‘see the person you are talking to, and for them to see you’
  - in both social and business contexts
  - except for novelty/gimmicky purposes (but on reflection, not felt to be sustained usage)
- Overriding perceptions are:
  - you generally know what the person looks likes
  - it is not *absolutely* necessary to see the person you are talking to
  - audio will suffice in the majority of circumstances
- Barriers to take-up also reinforced by price and quality perceptions
  - expect service to be expensive
  - expect quality to be poor
- Whilst the majority have little interest in seeing the other person, for some, there is an interest in being able to see:
  - the surroundings/ environment where the other person is
  - or an object/artefact the other person has with them

# Rel 99 GSM/WCDMA Network & 2-Way Video Communication

- 2-Way Video communication is a real time service requiring **high bandwidth** with managed **Quality of Service** for the duration of the call
- In early phases, this has to be **delivered in dual mode GSM/WCDMA Rel 99 network environment** I.e. "islands" of WCDMA coverage, but seamless services provision will rely on GSM/GPRS + WCDMA circuit & packet capabilities:
  - 2.5G circuit bearers
    - available CSD = 9.6/14.4kbps (HSCSD not widely deployed)
    - QoS guaranteed in CS environment
  - 2.5G packet bearers
    - practical PS data rate ~20kbps
    - best effort QoS
- **Lack of dynamic bearer rate negotiation in Rel 99** environment means available data rate for real time 2-way video communication in both GSM and WCDMA coverage areas will be constrained by the lowest GSM bearer rate in the network
- Nokia Mobile Phones believe 2-Way Video Communication becomes commercially viable in Rel 4/5 environment
  - based on packet bearers (higher available data rate in GSM footprint)
  - dynamic bearer rate negotiation
  - QoS support enables real time services on packet



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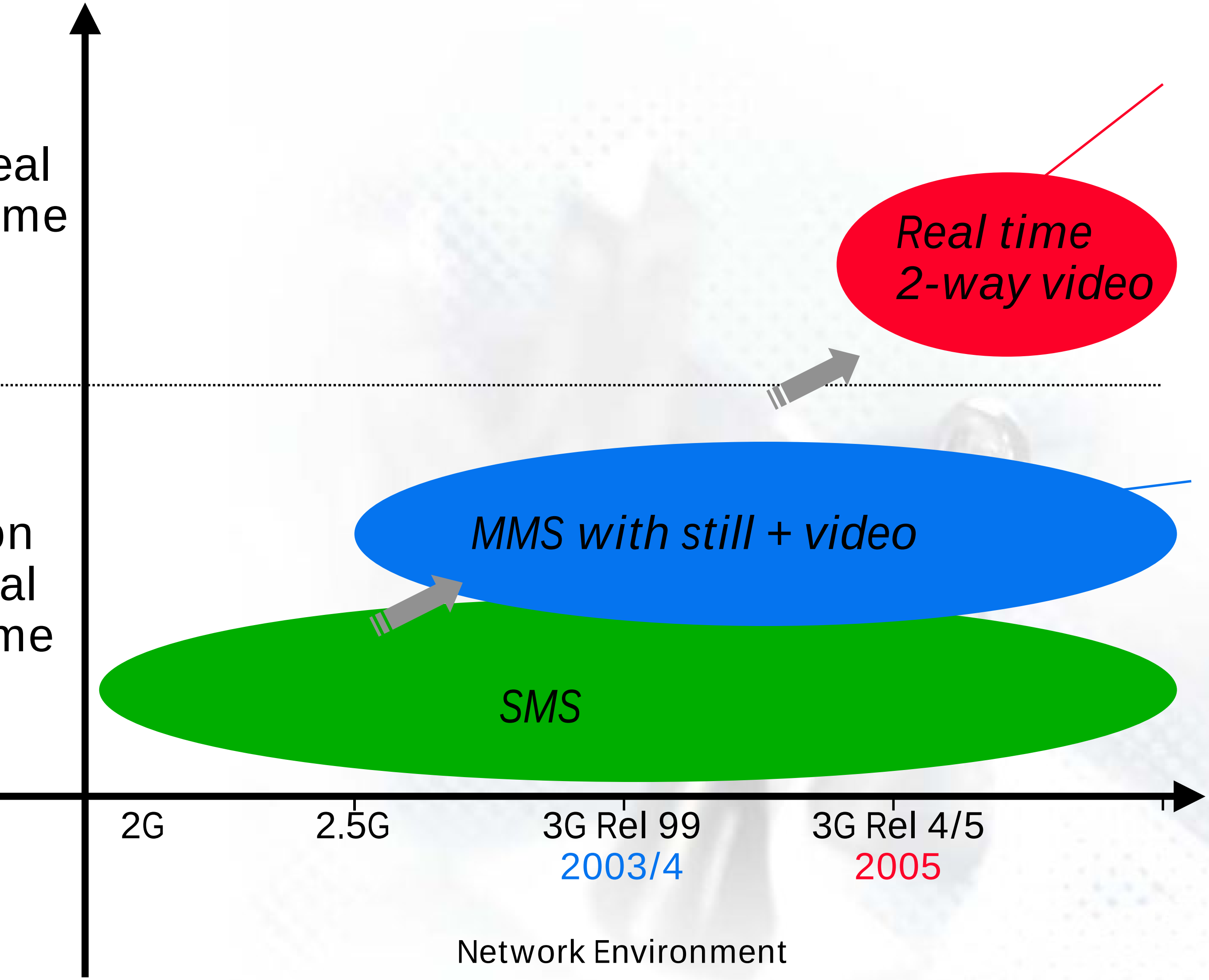
# Summary of Vision for Mobile P2P Video Communications

The ability to ***capture and share experiences visually***, whatever the place, whatever the time will ***fundamentally change*** the way people ***communicate***




**Evolves to 2-way video on pacl bearers in Rel4/5 network environment**

- Dynamic bearer rate negotiatior
- QoS with packet based real time service devliery

**P2P Video happens first with MMS**

- High appeal across broad selection of consumer segment
- Builds consumer awareness & understanding of Mobile Video communications



NOKIA

# Packet-Based Video Communications

- Packet based video communications can be built on a radio interface agnostic basis - they are not tied to a particular WCDMA bearer. They can be implemented in a number of radio access technologies but are ideally suited to WCDMA and EDGE. This broadens the potential number of subscribers who can be connected and enables a truly global service to be offered.
- Services based on IMS (IP Multimedia Subsystem) using SIP call control enable an extremely flexible approach to video communication to enable:
  - Set-up of video communications during an existing voice call
  - Creation of one way, as well as two way video communication streams. This enables more extensive use cases such as seeing the surroundings of the calling party rather than his face.
  - Removal and modification of the video call under the users' control.
- Packet-based solutions also enable easier inter-working with fixed internet clients as the same packet-based protocols are in use.
- Release 4 and 5 offer better support for end to end quality of service.

# SMIL – its role in Streaming Media

- SMIL
  - Synchronised Multimedia Integration Language
  - Defacto for presentation of streaming media
  - SMIL is to Streaming what HTML is to web-browsing
  - Streaming player can incorporate SMIL
  - SMIL can take various media components from various sources and control playback of media components in intended way (sequence, timing, overlays, layouts)

# Video and Audio Tools

Converters

- Multimedia Converter 2.0
- 3GP to AVI converter: http://miksoft.evzonastravel.com/mobile3GPconverter.htm
- Xilisoft Video Converter

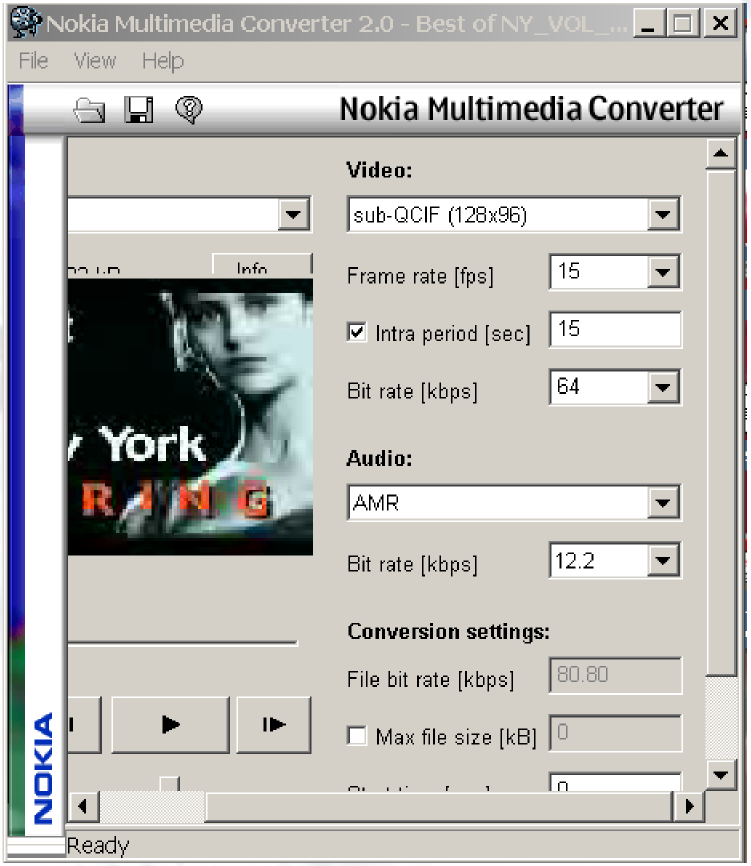
  http://www.xilisoft.com/video-converter.html

Players

- Apple Quicktime player version 6.4 and above
- Real Player
- Nokia Multimedia Player

# Nokia Multimedia Converter *2.0

- Support for 128kbps bit stream
- Converts AVI, WAV, MPEG, MP3
- Wideband and narrowband AMR supported.



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# Mobile TV

# Mobile TV

## MPEG-2 over DVB-T

24 Mbps / MUX

4-5 Mbps/program

3 - 5 TV channels
for large screen

**In-building coverage**

**Power saving**

**Optimal capacity**

## IP Datacast over DVB-H

16 QAM / 11 Mbps/ MUX

QPSK / 5 Mbps / MUX


200 - 500 kbps/program

10 – 55 channels
for small screen

# IP Datacasting

| | Mobility | Datarate | Frequency band | Comments |
|---|---|---|---|---|
| DVB-H | High | 5 – 11 Mbits/s | UHF<br>(470 – 702 MHz) | |
| DAB | High | Max 1,15 Mbit/s | VHF | VHF antenna size? |
| Korean<br>Terrestrial DMB | High | Max 1,15 Mbit/s | VHF | VHF antenna size? |
| ISDB-T | High<br>(one segment) | Max 1,5 Mbit/s<br>Typical 800 kbit/s | VHF + UHF<br>(91 – 770 MHz) | Japan only |
| Chinese Standards | | | | Under evaluation |
| 3GPP MBMS (Rel 6) | High | 300 kbit/s<br>(multicast) | WCDMA<br>(1920 – 1980 MHz + 2110 – 2170 MHz) | Limited capacity and medium cell size |
| 3GPP2 BCMCS | High | For 1X: 384 kbps (Max)<br>For 1xEV-DO: 204-409 kbps (typical) | All CDMA bands possible, e.g: 800MHz, 1700MHz,1900MHz, 2GHz | Refer to two papers (in IEEE Comm Mag, Feb 2004) |
| WLAN | Low | Max 11 Mbit/s | 2,4 GHz | Small cell size |
| RDS | High | ~100 bit/s | FM – Radio band<br>~100 MHz | Too low bitrate |

# Visual Radio

# Visual Radio

Unmodified traditional FM broadcast — FM

| | | | | | | |
|---|---|---|---|---|---|---|
| music | news | ad | music | ad | music | local events |

FM Tuner


New interactive parallel channel — GPRS

| | | | | | | |
|---|---|---|---|---|---|---|
| music info + quiz | more details | ad | music info + purchase | ad | music info + rating | extra info |

Visual Radio Application

# Visual Radio System Structure